

Functions and relations

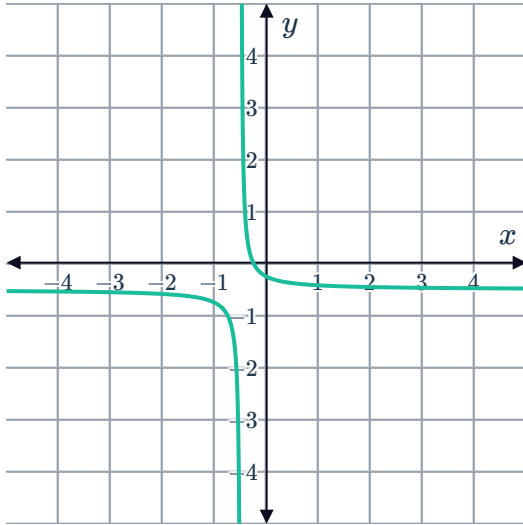


- 1 Express the relation $\{(2, 2), (4, 4), (6, 3), (7, 5)\}$ in a table of values.
- 2 Express the relation $\{(1, 2), (3, 5), (7, 8), (8, 11)\}$ as a series of points on the coordinate plane.
- 3 Determine whether the following statements are true or false.
 - a When working with a function, substituting a certain value of x into the formula gives only one value of y for that value of x .
 - b All relations are functions.
 - c Some relations are functions.
 - d All functions are relations.
 - e No functions are relations.
 - f No relations are functions.
 - g A relation always passes the vertical line test.
 - h The graph of every straight line is a function.
 - i The graph of every non-vertical straight line is a function.
 - j There is no straight line graph that is a function.
 - k The graph of every straight line is a relation.
 - l The graph of every non-horizontal straight line is a function.
- 4 Identify the name used to describe a graph where for some value of x , there exists two or more different values of y .

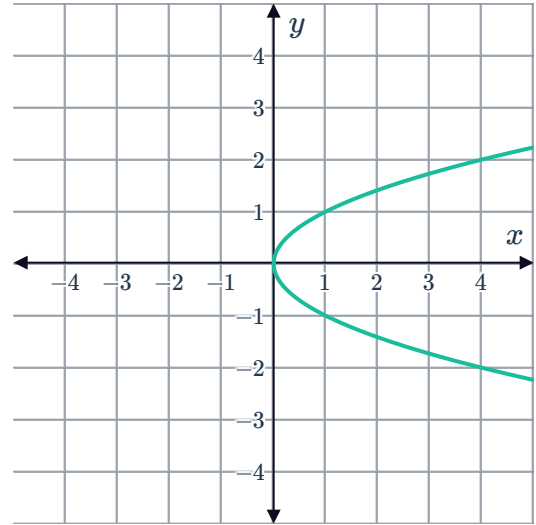
- 5 Determine whether each of the following graphs describes a relation only or both a relation and a function.



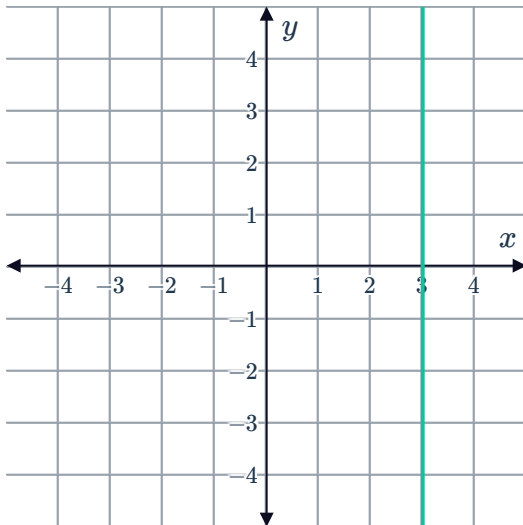
a



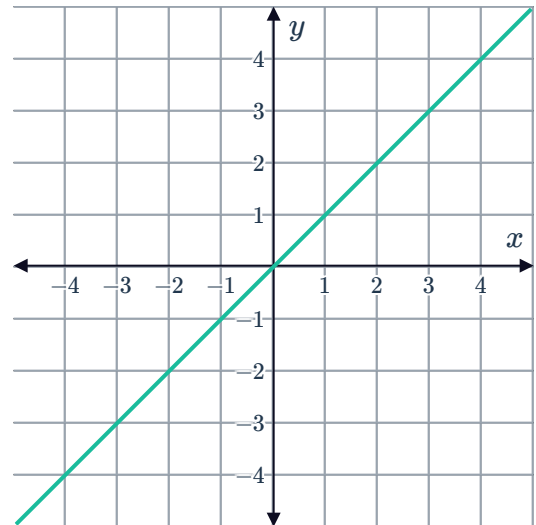
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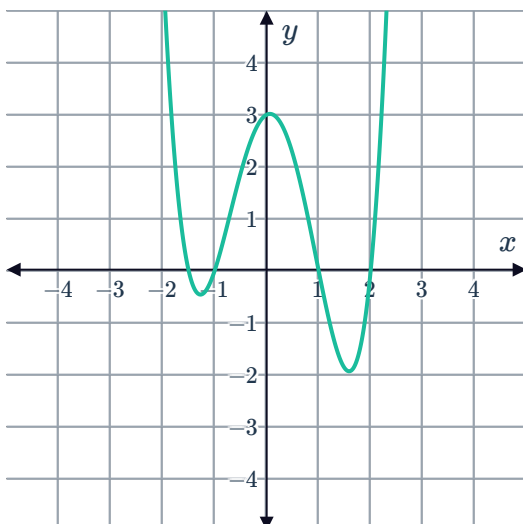
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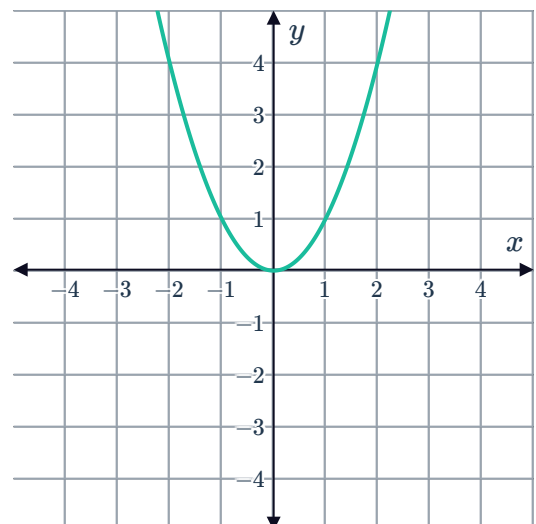
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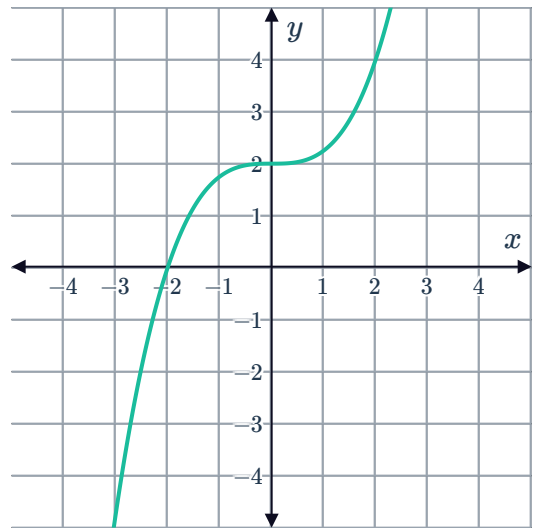
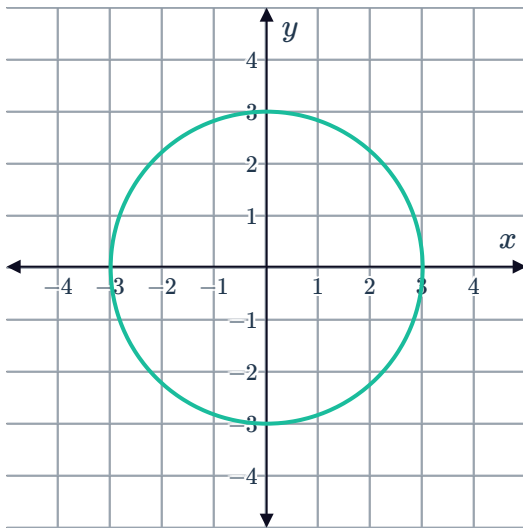
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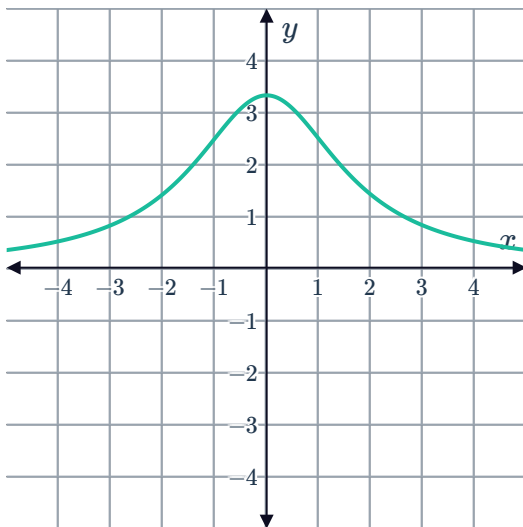
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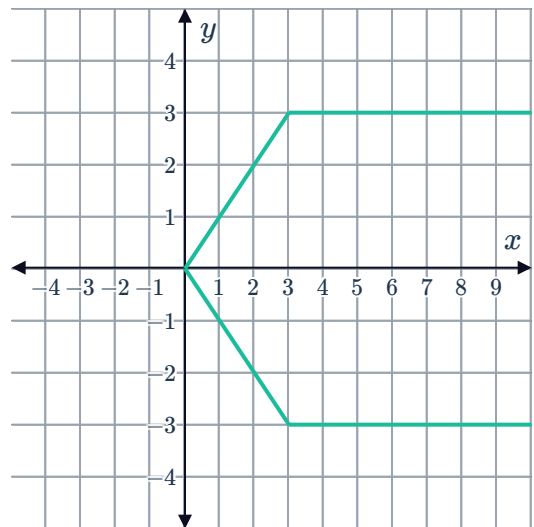
g



i



j



6 Consider the following set of points:
 $\{(16, 14), (8, 28), (-15, 14), (-20, -28)\}$

- a List the domain of the set of points.
- b List the range of the set of points.
- c State whether the points describe a function.

7 State whether the following set of points describe a function, a relation, both, or neither.

- a $\{(2, 5), (7, -3), (5, 2), (-4, -9)\}$
- b $\{(2, 5), (2, 7), (-3, -4), (-9, 13)\}$
- c $\{(1, 5), (1, 1), (7, -2), (-5, -10)\}$
- d $\{(1, 5), (7, -2), (-5, -10), (13, -13)\}$
- e $\{(1, 5), (1, 7), (-2, -5), (-5, -10)\}$



- 8 The pairs of values in the table represent a relation between x and y . Determine whether they represent a function.

a

x	-4	-3	-2	-1	0	1	2	3	4
y	-4	-3	-2	-1	0	-1	-2	-3	-4

b

x	0	1	4	8	9	12	16	18	20
y	0	1	2	$2\sqrt{2}$	3	$2\sqrt{3}$	4	$3\sqrt{2}$	$2\sqrt{5}$

c

x	-9	-7	-6	-5	-3	-2	3	5	10
y	10	10	10	10	10	10	10	10	10

d

x	-8	-7	-6	-3	2	7	9	9	10
y	8	13	-18	-16	-15	-2	-4	11	-9

- 9 Consider the points in the table.

x	-4	-3	0	-1	0	1	2	3	4
y	4	3	2	1	0	1	2	3	4

- a Plot the points on the coordinate plane.
b State whether the relation is a function.

- 10 A relation is defined as follows:
 $y = 1$ if x is positive and $y = -1$ if x is 0 or negative.

- a Complete the following table.

x	-4	-3	-2	-1	0	1	2	3	4
y									

- b Plot the points on the coordinate plane.
c State whether the relation is a function.



11 Dodgi Mobile charges \$2.90 a minute plus a connection fee of \$0.60 for an international call.

a Complete the table.

Call length (minutes)	International Call cost
1	
2	
3	
4	
5	

b State whether the relation is a function.

12 A particular store is offering one free T-shirt for every two T-shirts purchased, where each T-shirt costs \$19.

a Complete the table.

Number of T-shirts	Total Cost (dollars)
1	
2	
3	
4	
5	

b State whether the relation is a function.

13 Consider these ordered pairs:

$\{(-9, -5), (-5, -10), (-5, -4), (-3, 7), (-2, -4), (-1, 1)\}$

a Plot the ordered pairs on the number plane.

b State whether the relation is a function.

c Identify the ordered pair needed to be removed from the set so that the remaining ordered pairs represent a function.



14 Tyrah makes scarfs to sell at the market. It costs her \$2 to produce each one, and she sells them for \$5.

- a** Consider when 1, 2, 3, 4 and 5 scarfs are sold. Graph the points representing the relation between the number of scarfs she manages to sell and her total profit.
- b** State whether the relation is a function.

15 Find the possible values of k such that each of the following relations does not represent a function.

a $\{(2, 1), (7, k), (5, 8), (3, 9)\}$

b $\{(6, 2), (8, 5), (1, 7), (k, 4)\}$

16 Determine whether each of the following table of values represents a function.

a

x	y
-4	5
2	4
1	2
1	1

b

x	y
3	-4
2	-3
1	-1
2	0

c

x	y
2	1
2	3
2	5
2	7

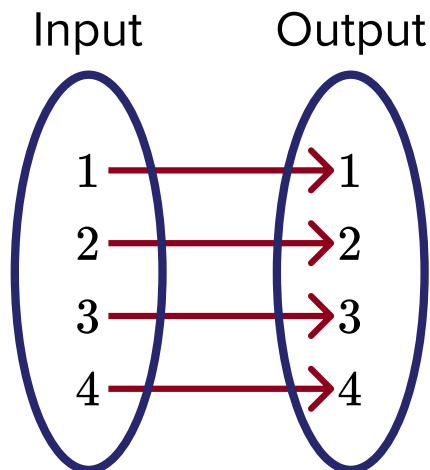
d

x	y
-1	2
0	1
1	2
2	3

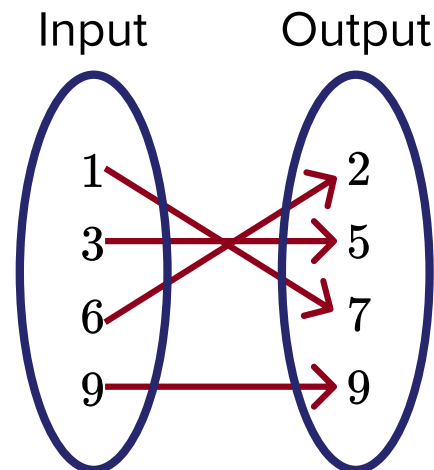
Additional questions

17 State whether each of the following mapping diagrams represents a function.

a



b



c

